

Practice Intermediate Algebra Test II (Version 1)

Name
Date
Course

I) Write your name, the date, and the course title (i.e. Intermediate Algebra).

II) Show all work.

III) If a problem requires algebraic steps to solve, draw a rectangle around your final answer to distinguish it from the other work.

IV) Read carefully. Make sure that you answer only what each question asks.

V) Each problem is worth four points for a total of 100 points.

① Graph the following quadratic expression. Write the coordinates of the vertex and two other points (one on each side of the line of symmetry).

$$y = -(x-3)(x+1)$$

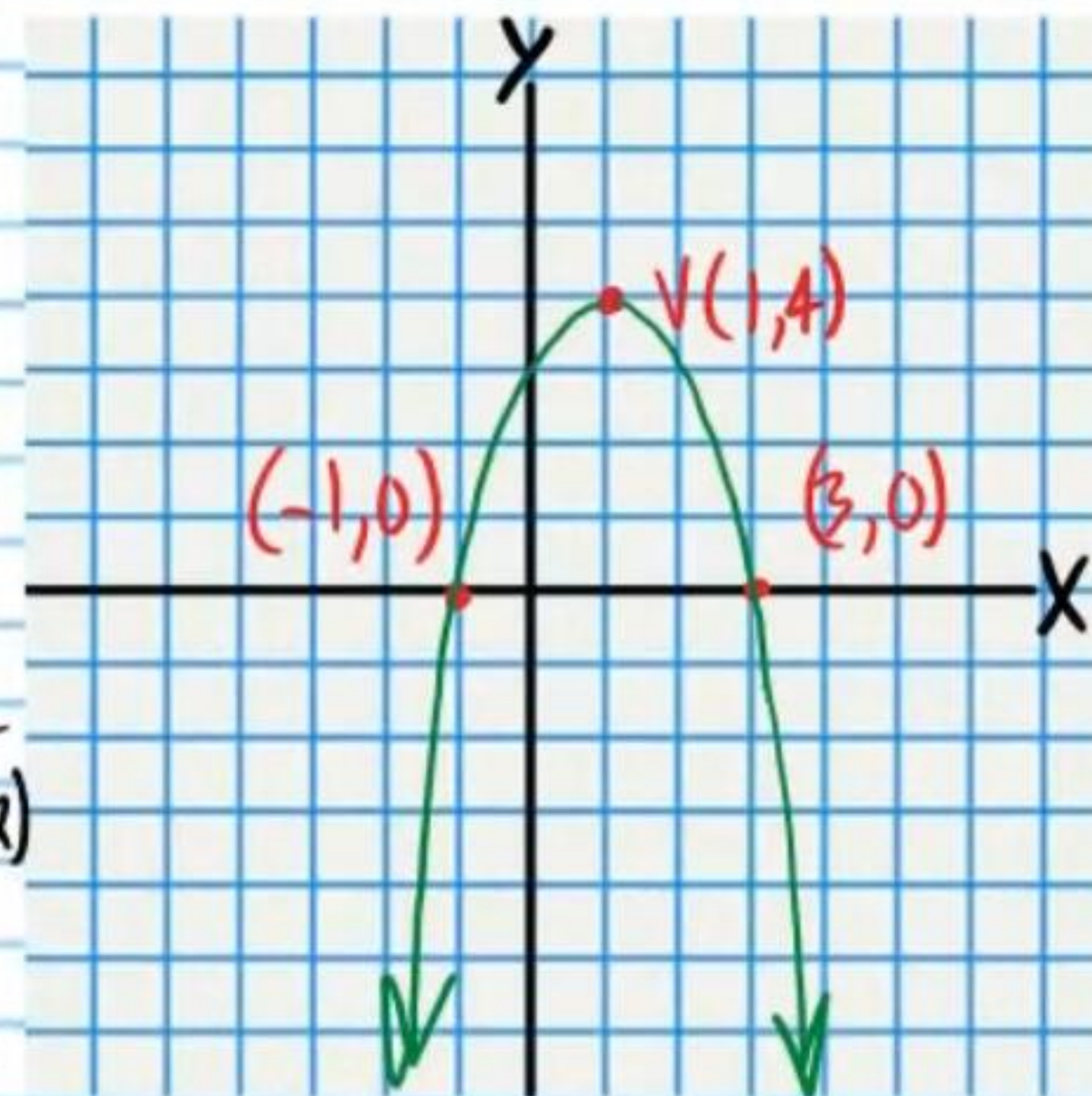
$$y = a(x-m)(x-h)$$

$$a = -1, m = 3, h = -1$$

$$V_x = \frac{m+h}{2} = \frac{3+(-1)}{2} = 1$$

$$V_y = -(1-3)(1+1) = -(-2)(2) = 4$$

$V(1,4)$



② Find the coordinates of the focus of the parabola in problem #1.

$$p = \frac{1}{4a} = \frac{1}{4(-1)} = -\frac{1}{4}$$

$$F(V_x, V_y + p)$$

$$F(1, 4 + (-\frac{1}{4}))$$

$$\boxed{F(1, 3\frac{3}{4})}$$

③ Find the equations of the line of symmetry and the directrix of the parabola in problem #1.

$$i) x = V_x$$

$$\boxed{x = 1}$$

$$ii) y = V_y - p$$

$$y = 4 - (-\frac{1}{4})$$

$$y = 4 + \frac{1}{4}$$

$$\boxed{y = 4\frac{1}{4}}$$

④ Graph the following quadratic expression. Write the coordinates of the vertex and two other points (one on each side of the line of symmetry).

$$y = 2x^2 + 20x + 50$$

$$y = ax^2 + bx + c$$

$$a = 2, b = 20, c = 50$$

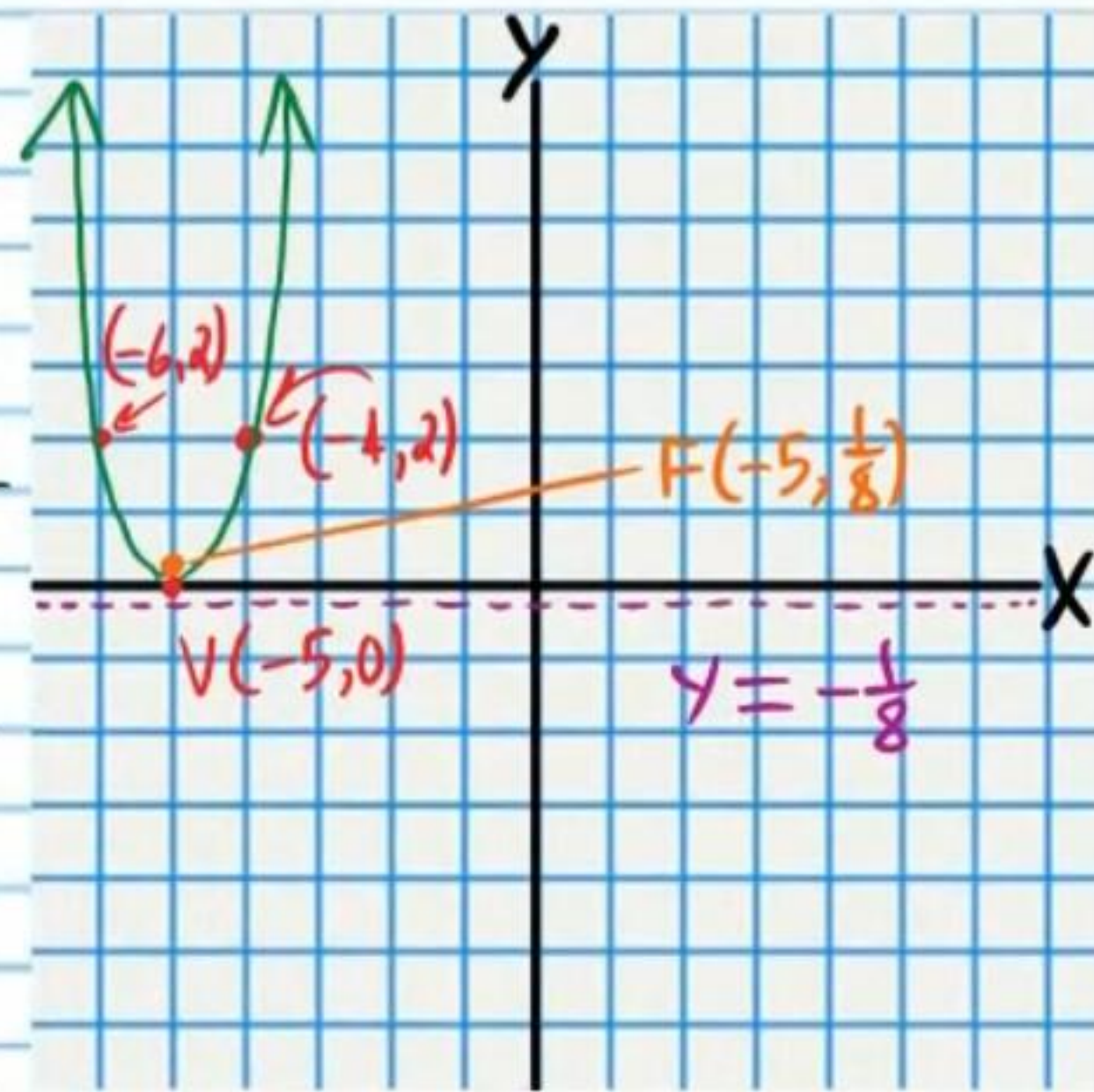
$$V_x = -\frac{b}{2a} = -\frac{20}{2(2)} = -5$$

$$V_y = 2(-5)^2 + 20(-5) + 50$$

$$V_y = 2(25) - 100 + 50$$

$$V_y = 0$$

$$V(-5, 0)$$



⑤ Find the coordinates of the focus and the equation of the directrix in problem #4 and draw them on the graph (use a dotted line for the directrix).

$$P = \frac{1}{4a} = \frac{1}{4(2)} = \frac{1}{8}$$

$$F(V_x, V_y + P)$$

$$F(-5, 0 + \frac{1}{8})$$

$$F(-5, \frac{1}{8})$$

$$y = V_y - P$$

$$y = 0 - \frac{1}{8}$$

$$y = -\frac{1}{8}$$

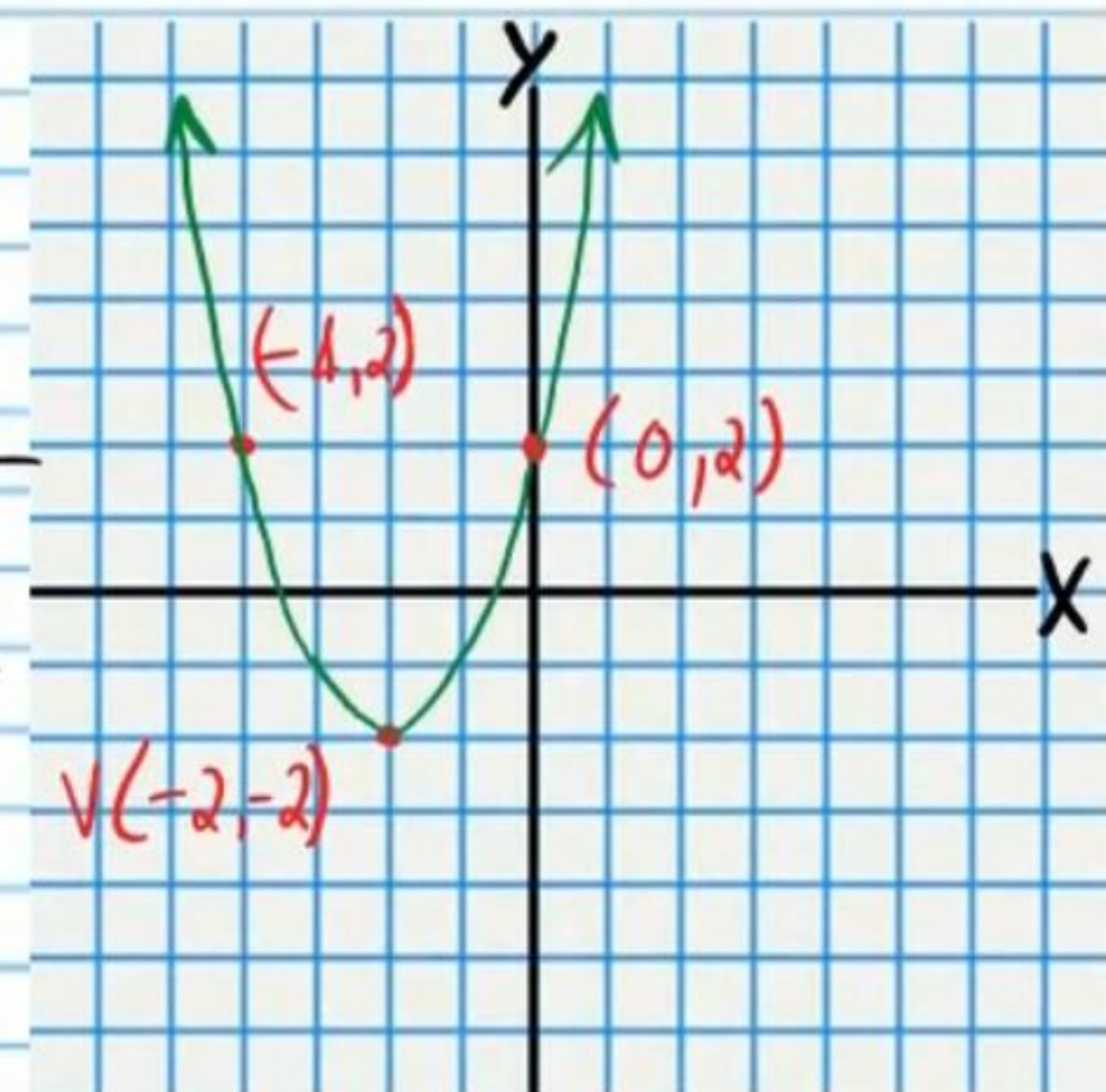
⑥ Graph the following quadratic expression. Write the coordinates of the vertex and two other points (one on each side of the line of symmetry).

$$(x+2)^2 - 2 = y$$

$$a(x-h)^2 + k = y$$

$$a = 1, h = -2, k = -2$$

$$V(h, k) = V(-2, -2)$$



Graph the following quadratic expression. Write the coordinates of the vertex, y-intercept, and x-intercepts on the graph.

⑦ $y = -\frac{1}{4}(x-4)^2 + 1$

$$y = a(x-h)^2 + k$$

$$a = -\frac{1}{4}, h = 4, k = 1$$

$$V(h, k) = V(4, 1)$$

$$x = 0$$

$$y = -\frac{1}{4}(0-4)^2 + 1$$

$$y = -3$$

$$(0, -3)$$

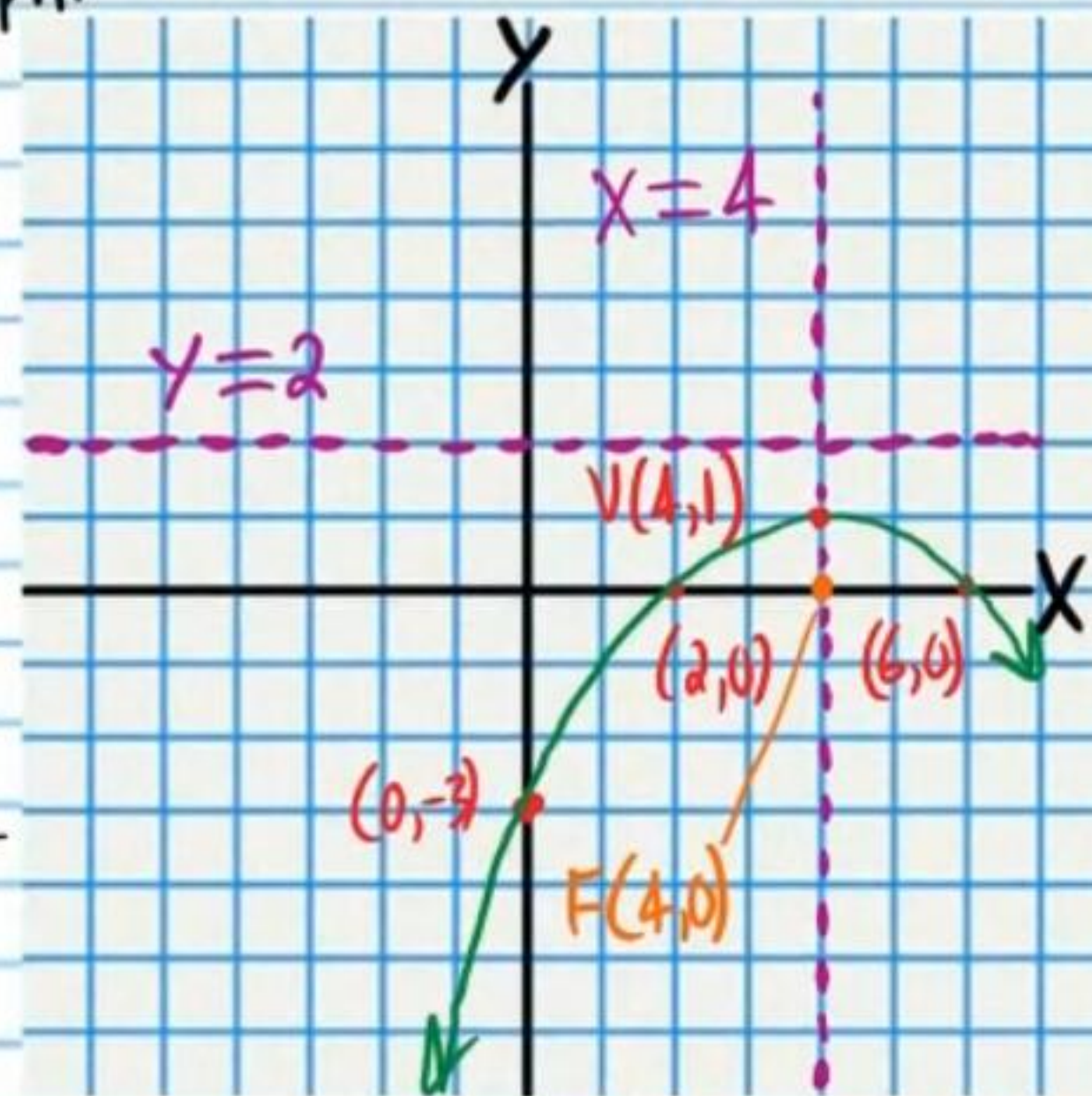
$$y = 0$$

$$0 = -\frac{1}{4}(x-4)^2 + 1$$

$$-1 = -\frac{1}{4}(x-4)^2$$

$$4 = (x-4)^2$$

$$\pm 2 = x - 4$$



$$2 = x - 4 \text{ or } -2 = x - 4$$

$$x = 6 \text{ or } x = 2$$

⑧ Draw the directrix of the parabola in problem #7 on the graph (as a dotted line) and write its equation.

$$P = \frac{1}{4a} = \frac{1}{4(-\frac{1}{4})} = -1$$

$$y = V_y - P$$

$$y = 1 - (-1)$$

$$y = 1 + 1$$

$$\boxed{y = 2}$$

⑨ Find the coordinates of the focus and the equation of the line of symmetry of the parabola in problem #7. Draw them (a point and a dotted line) on the graph.

i) $F(V_x, V_y + P)$

$$F(4, 1 + (-1))$$

$$\boxed{F(4, 0)}$$

ii) $x = V_x$

$$\boxed{x = 4}$$

⑩ Graph the following quadratic expression. Write the coordinates of the vertex and the x-intercepts.

$$-3x^2 - 8x - 4 = y$$

$$ax^2 + bx + c = y$$

$$a = -3, b = -8, c = -4$$

$$V_x = -\frac{b}{2a} = -\frac{(-8)}{2(-3)} = \frac{8}{-6}$$

$$= -\frac{4}{3}$$

$$= -1\frac{1}{3}$$

$$V_y = -3\left(-\frac{4}{3}\right)^2 - 8\left(-\frac{4}{3}\right) - 4$$

$$= -3\left(\frac{16}{9}\right) + \frac{32}{3} - 4$$

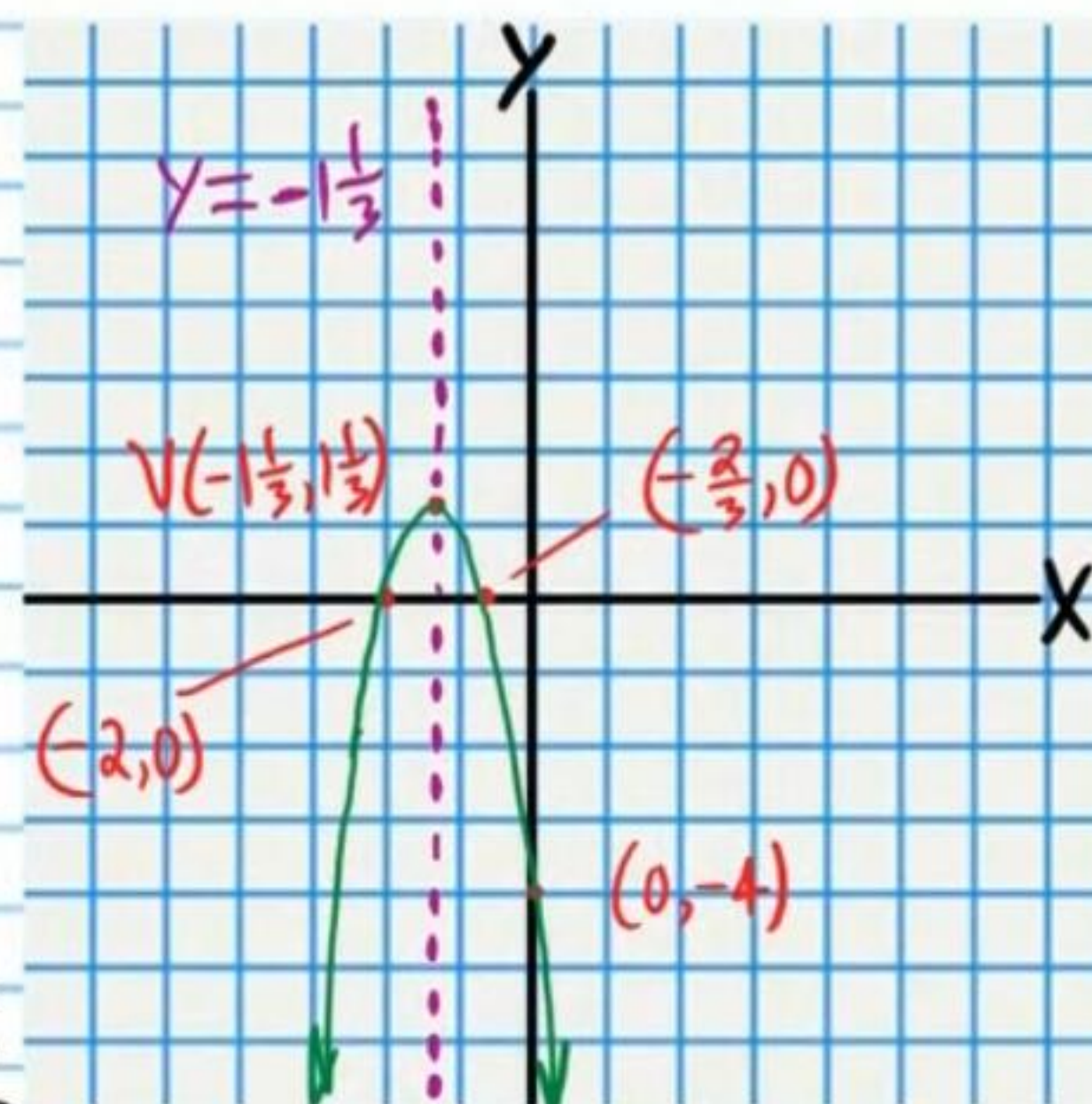
$$= -\frac{48}{9} + \frac{96}{9} - \frac{36}{9}$$

$$= \frac{12}{9}$$

$$= \frac{4}{3}$$

$$= 1\frac{1}{3}$$

$$V\left(-1\frac{1}{3}, 1\frac{1}{3}\right)$$



$$-3x^2 - 8x - 4 = 0$$

$$3x^2 + 8x + 4 = 0$$

$$(3x+2)(x+2) = 0$$

$$3x+2=0 \text{ or } x+2=0$$

$$x = -\frac{2}{3} \text{ or } x = -2$$

⑪ Draw the line of symmetry of the parabola in problem #10 on the graph and write its equation.

$$x = V_x$$

$$x = -1\frac{1}{3}$$

⑫ What is the y-intercept of the graph in problem #10?

$$(0, -4)$$

⑬ Find the discriminant of the quadratic expression below and use it to determine the number of x-intercepts of the corresponding graph.

$$y = 10x^2 - 3x + 1$$

$$y = ax^2 + bx + c$$

$$a = 10, b = -3, c = 1$$

$$D = b^2 - 4ac = (-3)^2 - 4(10)(1) = 9 - 40 = -31$$

No x-intercepts

⑭ Write an equation of the parabola. Hint: It is easiest to use vertex form.

$$y = a(x-h)^2 + k$$

$$V(-4, 3) \quad P(-1, 6)$$

$$6 = a(-1 - (-4))^2 + 3$$

$$6 = a(-1 + 4)^2 + 3$$

$$6 = a(3)^2 + 3$$

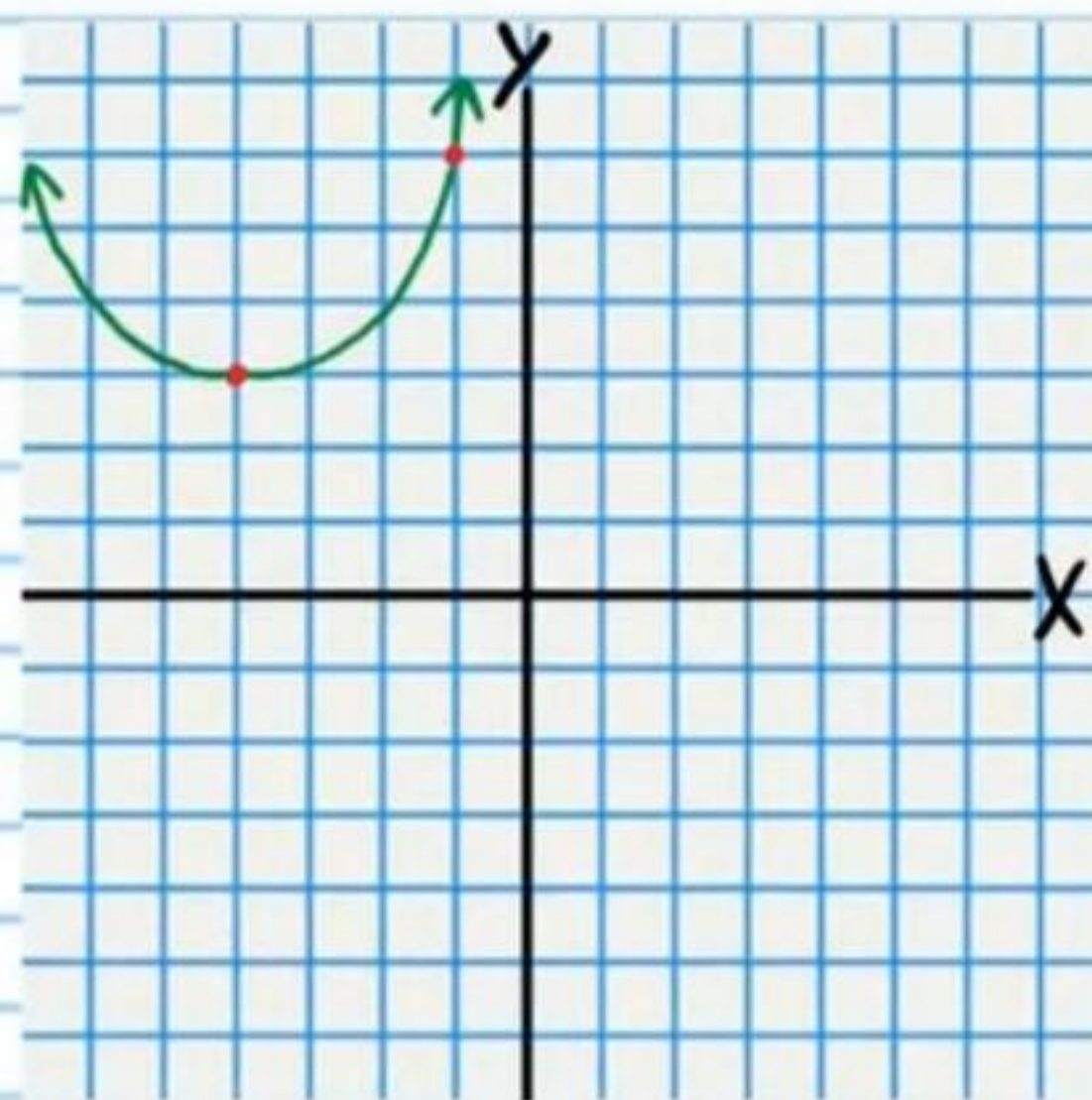
$$6 = a(9) + 3$$

$$3 = 9a$$

$$\frac{1}{3} = a$$

$$y = \frac{1}{3}(x - (-4))^2 + 3$$

$$y = \frac{1}{3}(x + 4)^2 + 3$$



For each problem below, write an equation of the parabola with the given characteristics. Hint: It is easiest to use vertex form. Assume that all corresponding parabolas open up or down.

⑮ Vertex (2, 5) and y-intercept -10.

$$V(2, 5) \quad P(0, -10)$$

$$y = a(x-h)^2 + k$$

$$-10 = a(0 - 2)^2 + 5$$

$$-10 = a(-2)^2 + 5$$

$$-10 = a(4) + 5$$

$$-15 = 4a$$

$$a = -\frac{15}{4}$$

$$y = -\frac{15}{4}(x - 2)^2 + 5$$

⑩ Focus $(4, -4\frac{3}{4})$ and directrix $y = -5\frac{1}{4}$.

$$F(V_x, V_y + P)$$

$$y = V_y - P$$

$$-4\frac{3}{4} = V_y + P$$

$$-5\frac{1}{4} = V_y - P$$

$$\begin{cases} -4\frac{3}{4} = V_y + P \\ -5\frac{1}{4} = V_y - P \end{cases}$$

$$\begin{aligned} -4\frac{3}{4} &= -5 + P \\ \frac{1}{4} &= P \end{aligned}$$

$$-10 = 2V_y$$

$$-5 = V_y$$

$$P = \frac{1}{4a}$$

$$\frac{1}{4} = \frac{1}{4a}$$

$$a = 1$$

$$V_x = F_x = 4$$

$$V(4, -5)$$

$$y = (x - 4)^2 - 5$$

⑪ Convert the equation below to intercept form.

$$y = -(x - 6)^2 + 9$$

$$y = -(x - 6)(x - 6) + 9$$

$$y = -(x^2 - 6x - 6x + 36) + 9$$

$$y = -(x^2 - 12x + 36) + 9$$

$$y = -x^2 + 12x - 36 + 9$$

$$y = -x^2 + 12x - 27$$

$$y = -(x^2 - 12x + 27)$$

$$y = -(x - 9)(x - 3)$$

⑫ Convert the equation below to vertex form.

$$2(x - 1)(x + 5) = y$$

$$2(x^2 + 5x - x - 5) = y$$

$$2(x^2 + 4x - 5) = y$$

$$2x^2 + 8x - 10 = y$$

$$x^2 + 4x - 5 = \frac{y}{2}$$

$$x^2 + 4x = \frac{y}{2} + 5$$

$$x^2 + 4x + 4 = \frac{y}{2} + 5 + 4$$

$$(x + 2)(x + 2) = \frac{y}{2} + 9$$

$$(x + 2)^2 = \frac{y}{2} + 9$$

$$(x + 2)^2 - 9 = \frac{y}{2}$$

$$2(x + 2)^2 - 18 = y$$

① Convert the equation below to the two other forms (i.e. intercept and vertex).

$$y = -3x^2 + 6x + 24$$

$$y = ax^2 + bx + c$$

$$a = -3, b = 6, c = 24$$

$$V_x = -\frac{b}{2a} = -\frac{6}{2(-3)} = 1$$

$$\begin{aligned} V_y &= -3(1)^2 + 6(1) + 24 \\ &= -3(1) + 6 + 24 \\ &= 27 \end{aligned}$$

$$V(1, 27)$$

$$y = -3(x-1)^2 + 27$$

$$y = -3x^2 + 6x + 24$$

$$y = -3(x^2 - 2x - 8)$$

$$y = -3(x-4)(x+2)$$

Graph the following quadratic expressions. Write the coordinates of the vertex, y-intercept, and focus on each graph. Draw the directrix (as a dotted line) and write its equation.

② $\frac{1}{2}x(x+4) = y$

$$\frac{1}{2}(x-0)(x+4) = y$$

$$a = \frac{1}{2}, m = 0, n = -4$$

$$V_x = \frac{m+n}{2} = \frac{0+(-4)}{2}$$

$$= -2$$

$$V_y = \frac{1}{2}(-2)(-2+4)$$

$$= -2$$

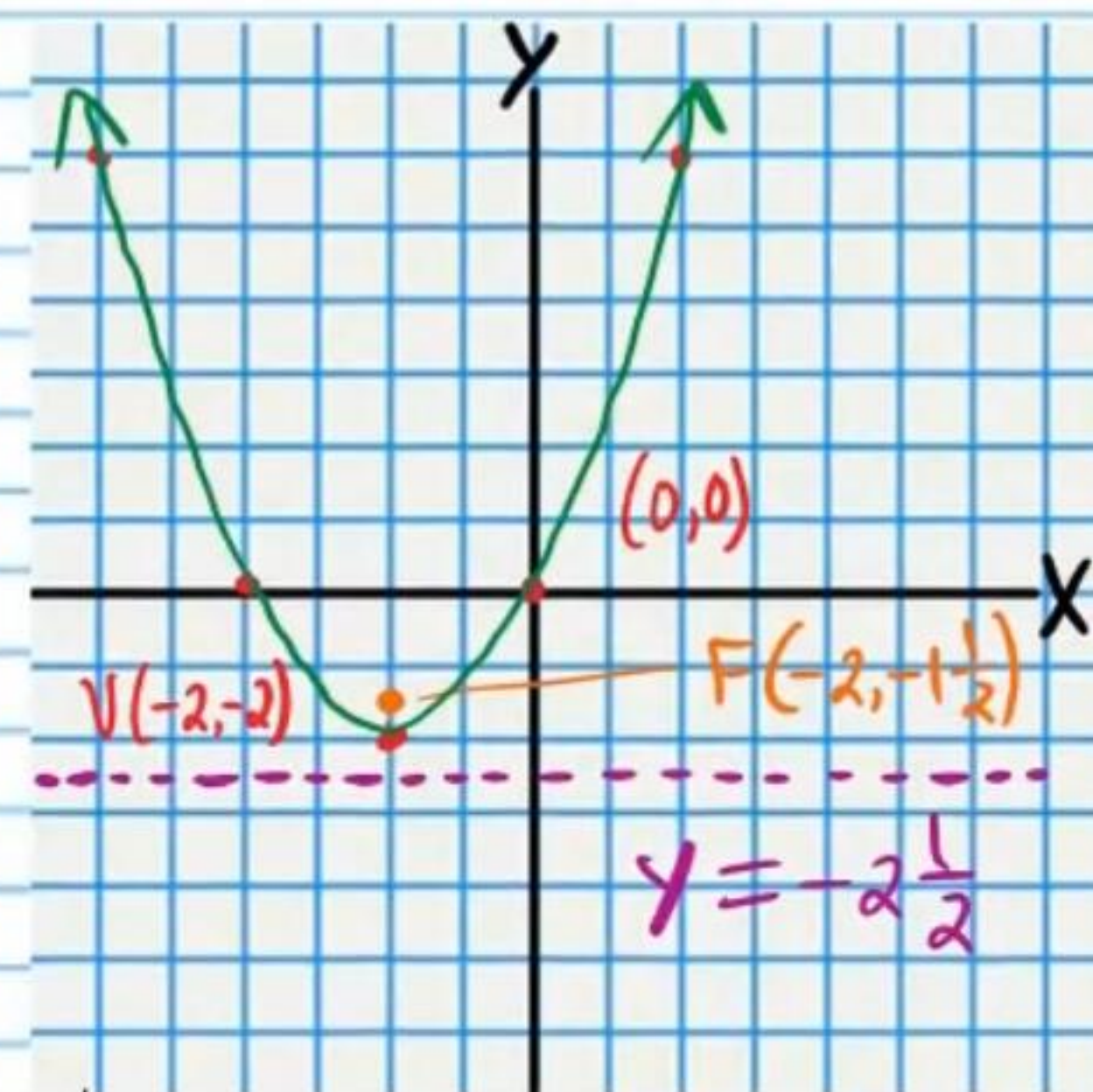
$$V(-2, -2)$$

$$p = \frac{1}{4a} = \frac{1}{4(\frac{1}{2})} = \frac{1}{2}$$

$$F(V_x, V_y + p)$$

$$F(-2, -2 + \frac{1}{2})$$

$$F(-2, -1\frac{1}{2})$$



$$y = V_y - p$$

$$y = -2 - \frac{1}{2}$$

$$y = -2\frac{1}{2}$$

21) $y = -\frac{1}{8}x^2 + x - 5$

$$y = ax^2 + bx + c$$

$$a = -\frac{1}{8}, b = 1, c = -5$$

$$V_x = -\frac{b}{2a} = -\frac{1}{2(-\frac{1}{8})}$$

$$= \frac{1}{-\frac{1}{4}}$$

$$= 4$$

$$V_y = -\frac{1}{8}(4)^2 + 4 - 5$$

$$V_y = -\frac{1}{8}(16) - 1$$

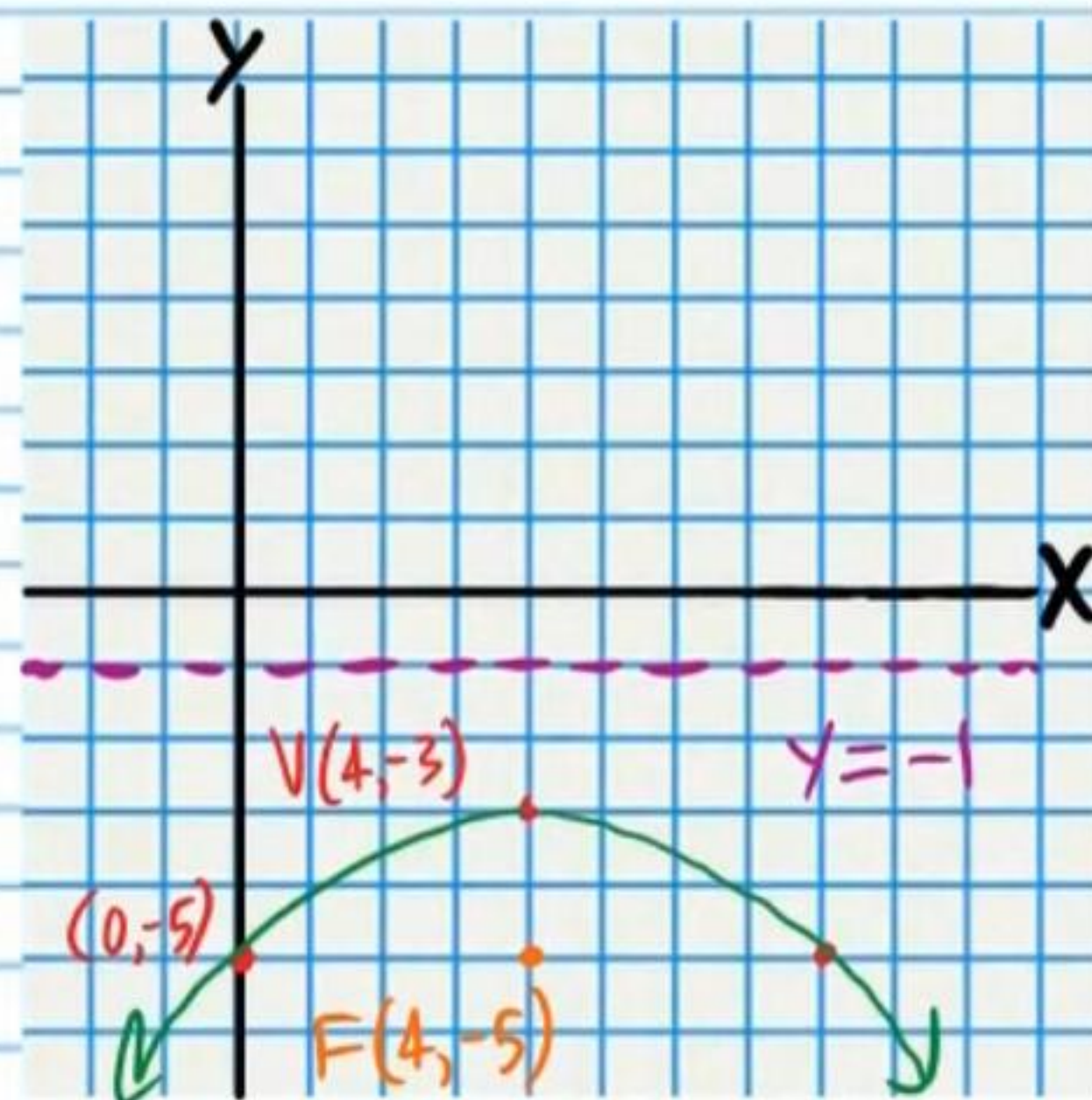
$$V_y = -3$$

$$V(4, -3)$$

$$P = \frac{1}{4a} = \frac{1}{4(-\frac{1}{8})} = \frac{1}{-\frac{1}{2}} = -2$$

$$F(V_x, V_y + P)$$

$$F(4, -3 + (-2)) = \boxed{F(4, -5)}$$



$$y = V_y - P$$

$$y = -3 - (-2)$$

$$y = -3 + 2$$

$$\boxed{y = -1}$$

22) Graph the following quadratic expression. Write the coordinates of the vertex, y-intercept, and x-intercepts.

$$y = \frac{1}{3}x^2 + 3$$

$$y = \frac{1}{3}(x-0)^2 + 3$$

$$y = a(x-h)^2 + k$$

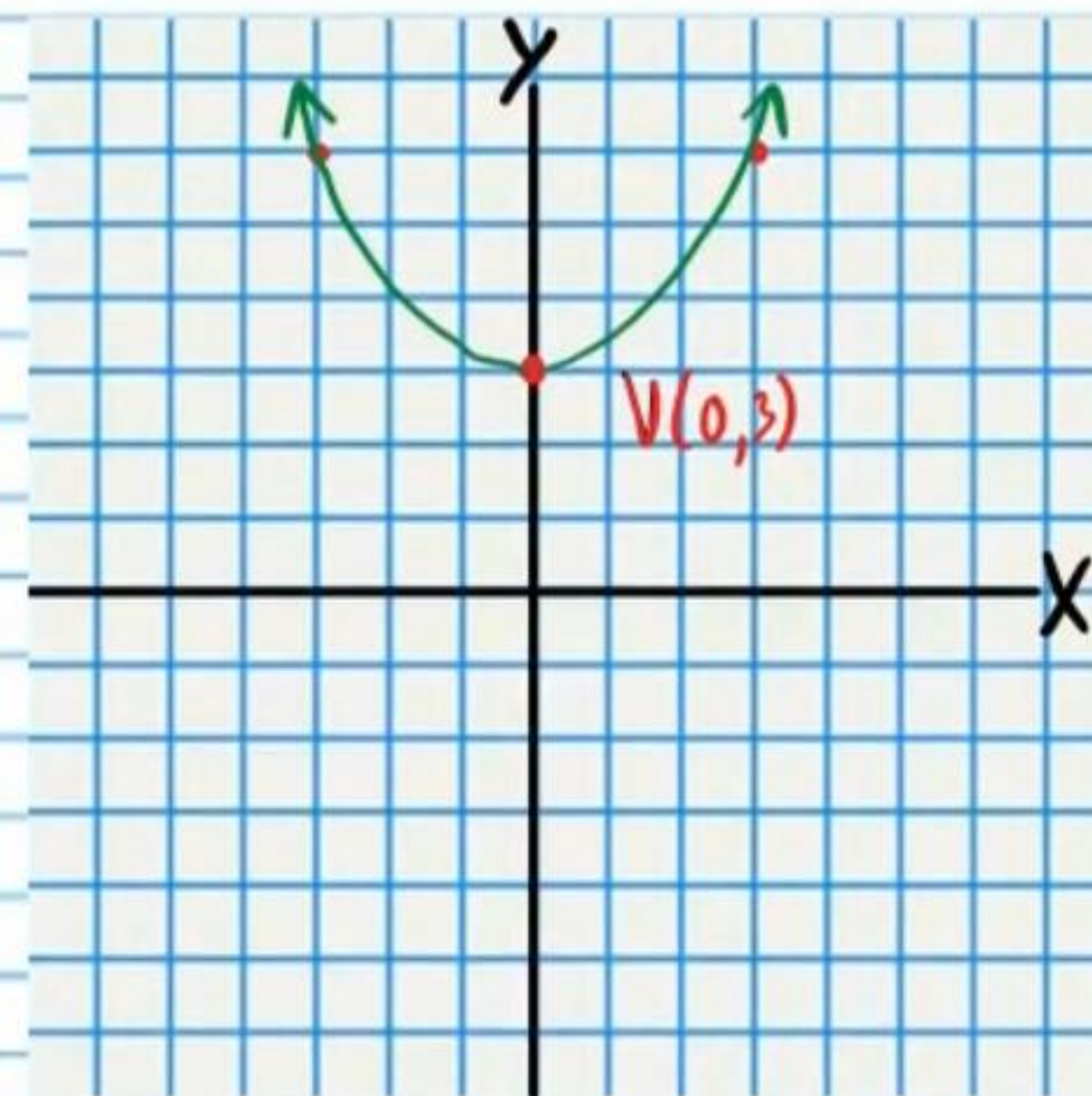
$$V(h, k) = V(0, 3)$$

$$0 = \frac{1}{3}x^2 + 3$$

$$-3 = \frac{1}{3}x^2$$

$$-9 = x^2$$

$\boxed{\text{No x-intercepts}}$



Graph the following quadratic expressions. Write the coordinates of the vertex and the x-intercepts on each graph. Write exact expressions for the x-intercepts. Draw the line of symmetry (as a dotted line) and write its equation.

(23) $y = 5(x+3)^2 - 10$

$$y = a(x-h)^2 + k$$

$$a = 5, h = -3, k = -10$$

$$V(h, k) = V(-3, -10)$$

$$y = 0$$

$$0 = 5(x+3)^2 - 10$$

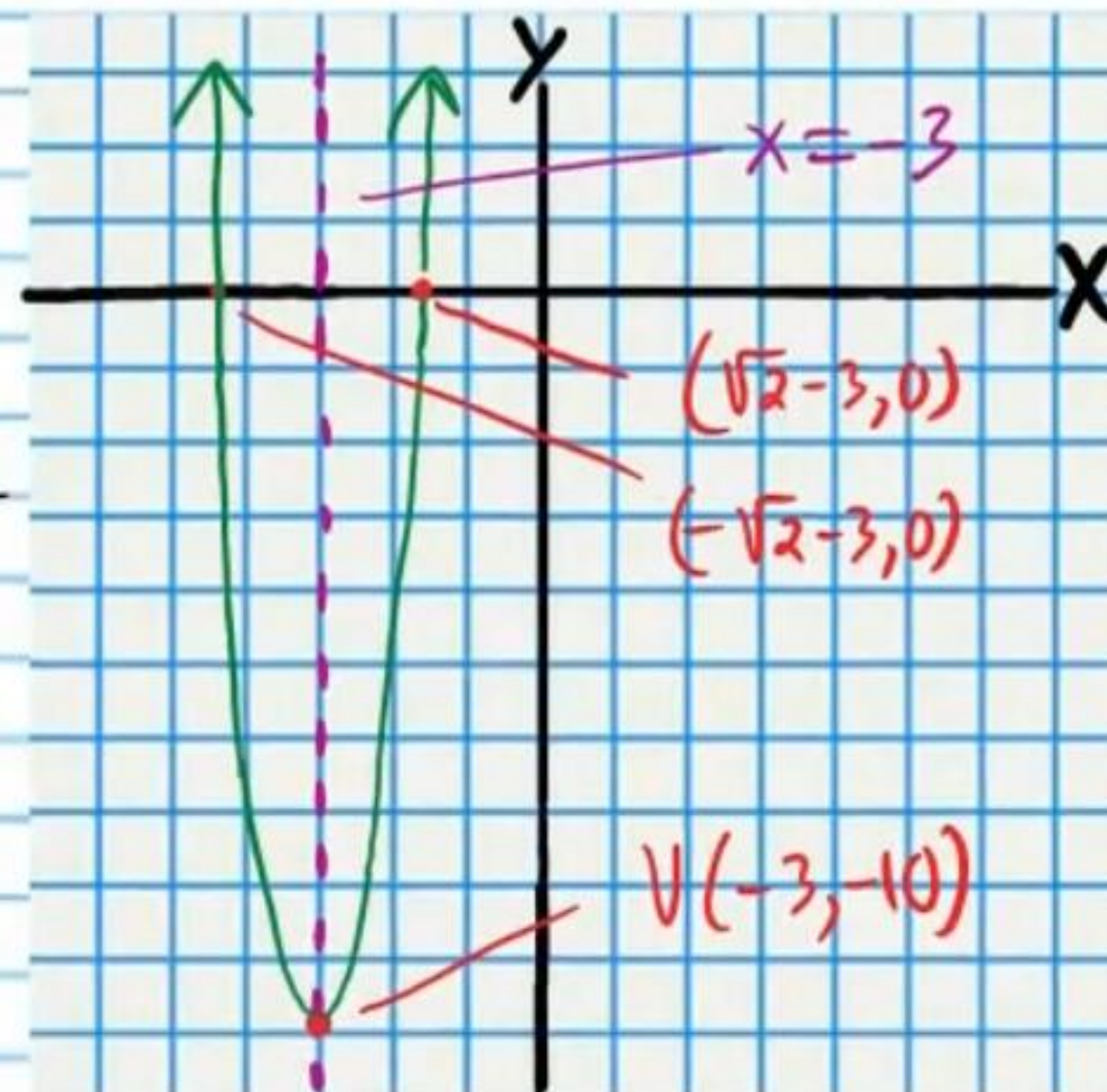
$$10 = 5(x+3)^2$$

$$2 = (x+3)^2$$

$$\pm\sqrt{2} = x+3$$

$$\sqrt{2} = x+3 \text{ or } -\sqrt{2} = x+3$$

$$\sqrt{2} - 3 = x \text{ or } -\sqrt{2} - 3 = x$$



$$\sqrt{1} < \sqrt{2} < \sqrt{4}$$

$$1 < \sqrt{2} < 2$$

(24) $-x^2 + 2x + 6 = y$

$$ax^2 + bx + c = y$$

$$a = -1, b = 2, c = 6$$

$$V_x = -\frac{b}{2a} = -\frac{2}{2(-1)} = 1$$

$$V_y = -(1)^2 + 2(1) + 6$$

$$V_y = -1 + 2 + 6$$

$$= 7$$

$$V(1, 7)$$

$$y = 0$$

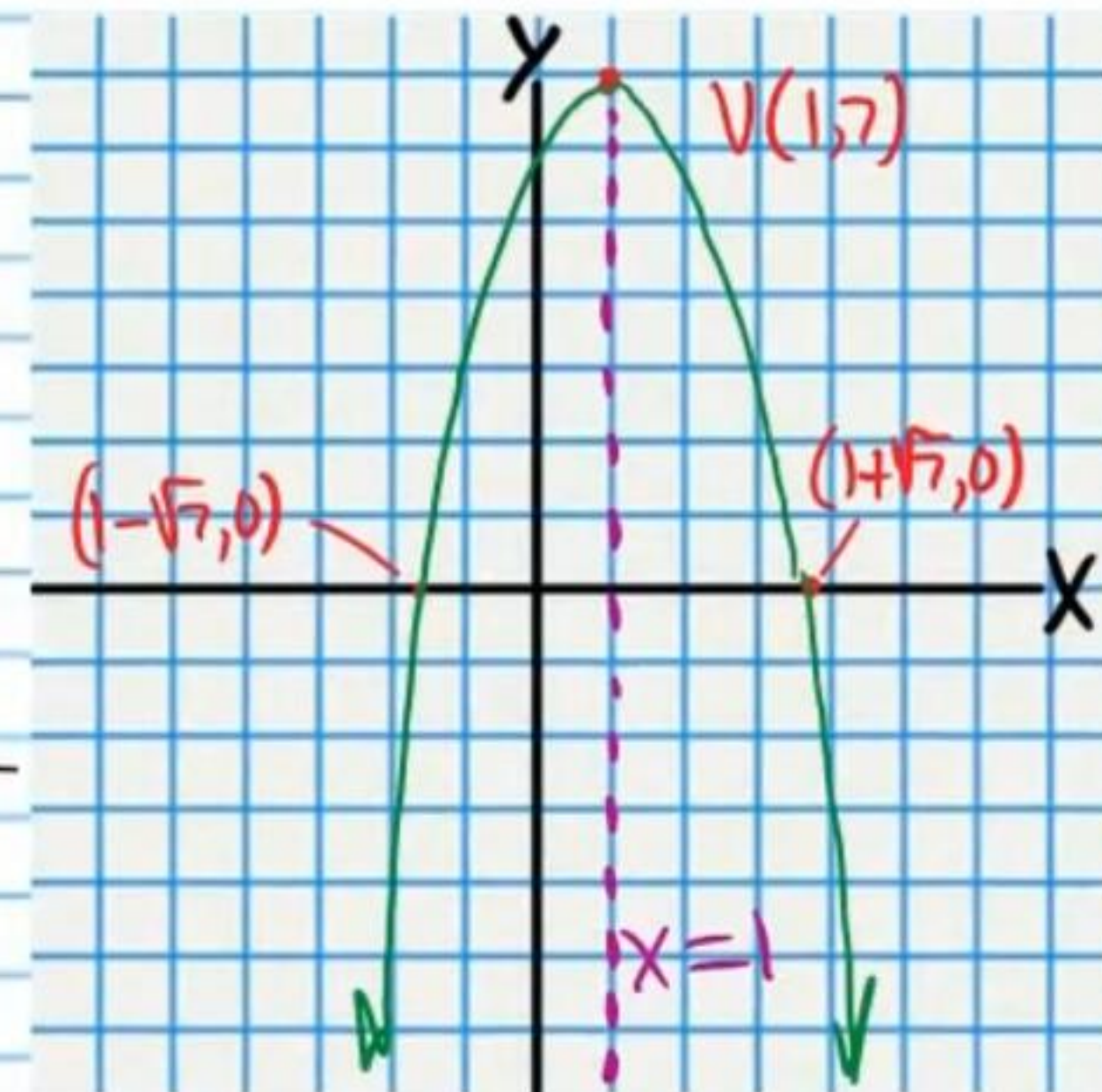
$$-x^2 + 2x + 6 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-2 \pm \sqrt{2^2 - 4(-1)(6)}}{2(-1)}$$

$$x = \frac{-2 \pm \sqrt{4 + 24}}{-2}$$

$$= \frac{-2 \pm \sqrt{28}}{-2} = \frac{-2 \pm 2\sqrt{7}}{-2} = 1 \pm \sqrt{7}$$



$$\sqrt{4} < \sqrt{7} < \sqrt{9}$$

$$2 < \sqrt{7} < 3$$

$$\approx 2.8$$

②5 Graph the following quadratic expression. Write the coordinates of the vertex, y-intercept, and x-intercepts on the graph. Write exact expressions for the x-intercepts.

$$2x^2 - 3x - 1 = y$$

$$ax^2 + bx + c = y$$

$$a = 2, b = -3, c = -1$$

$$V_x = -\frac{b}{2a} = -\frac{(-3)}{2(2)} = \frac{3}{4}$$

$$V_y = 2\left(\frac{3}{4}\right)^2 - 3\left(\frac{3}{4}\right) - 1$$

$$= 2\left(\frac{9}{16}\right) - \frac{9}{4} - 1$$

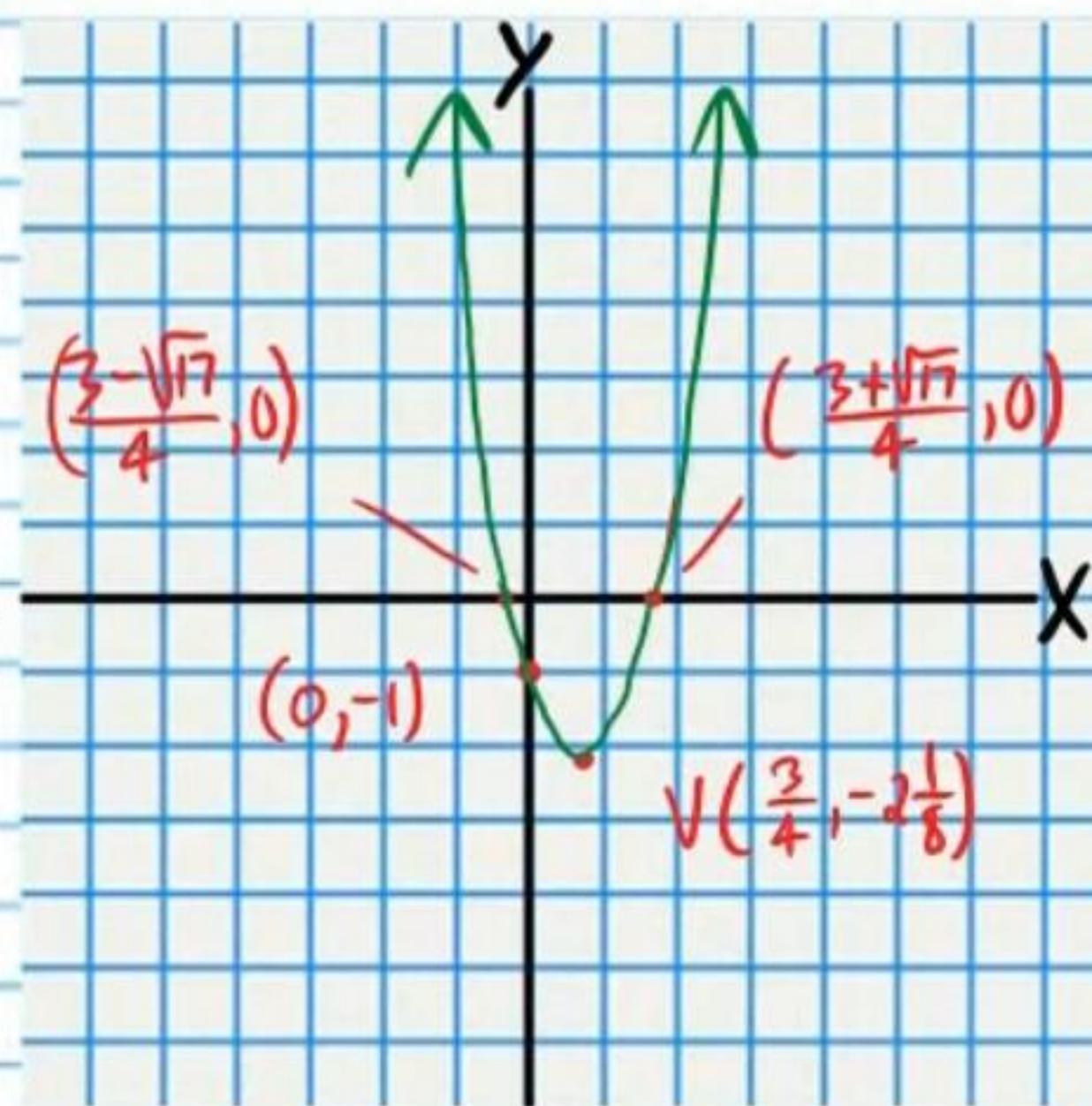
$$= \frac{18}{16} - \frac{36}{16} - \frac{16}{16}$$

$$= -\frac{34}{16}$$

$$= -2\frac{2}{16}$$

$$= -2\frac{1}{8}$$

$$V\left(\frac{3}{4}, -2\frac{1}{8}\right)$$



$$y = 0$$

$$2x^2 - 3x - 1 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-3) \pm \sqrt{(-3)^2 - 4(2)(-1)}}{2(2)}$$

$$x = \frac{3 \pm \sqrt{9+8}}{4}$$

$$x = \frac{3 \pm \sqrt{17}}{4}$$

$$\sqrt{16} < \sqrt{17} < \sqrt{25}$$

$$4 < \sqrt{17} < 5$$

$$\approx 4.1$$